

Theorem 6.2 : *If a line divides any two sides of a triangle in the same ratio, then the line is parallel to the third side.*

This theorem can be proved by taking a line DE such

that $\frac{AD}{DB} = \frac{AE}{EC}$ and assuming that DE is not parallel to BC (see Fig. 6.12).

If DE is not parallel to BC, draw a line DE' parallel to BC.

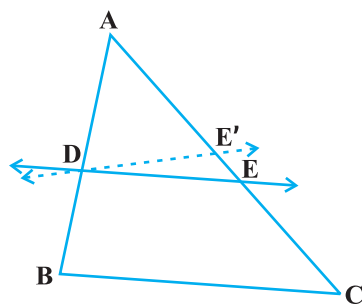


Fig. 6.12

So, $\frac{AD}{DB} = \frac{AE'}{E'C}$ (Why ?)

Therefore, $\frac{AE}{EC} = \frac{AE'}{E'C}$ (Why ?)

Adding 1 to both sides of above, you can see that E and E' must coincide. (Why ?)

Let us take some examples to illustrate the use of the above theorems.

Example 1 : If a line intersects sides AB and AC of a $\triangle ABC$ at D and E respectively

and is parallel to BC, prove that $\frac{AD}{AB} = \frac{AE}{AC}$ (see Fig. 6.13).

Solution :

DE $\parallel$ BC (Given)

So, $\frac{AD}{DB} = \frac{AE}{EC}$ (Theorem 6.1)

or, $\frac{DB}{AD} = \frac{EC}{AE}$

or, $\frac{DB}{AD} + 1 = \frac{EC}{AE} + 1$

or, $\frac{AB}{AD} = \frac{AC}{AE}$

So, $\frac{AD}{AB} = \frac{AE}{AC}$

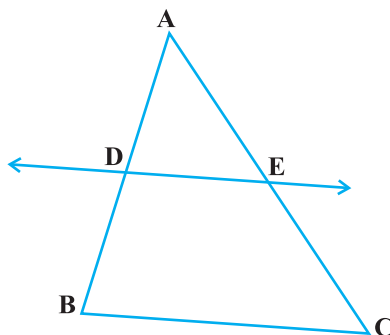


Fig. 6.13

Example 2 : ABCD is a trapezium with $AB \parallel DC$. E and F are points on non-parallel sides AD and BC respectively such that EF is parallel to AB

(see Fig. 6.14). Show that $\frac{AE}{ED} = \frac{BF}{FC}$.

Solution : Let us join AC to intersect EF at G (see Fig. 6.15).

$AB \parallel DC$ and $EF \parallel AB$ (Given)

So, $EF \parallel DC$ (Lines parallel to the same line are parallel to each other)

Now, in $\triangle ADC$,

$EG \parallel DC$ (As $EF \parallel DC$)

$$\text{So, } \frac{AE}{ED} = \frac{AG}{GC} \quad (\text{Theorem 6.1}) \quad (1)$$

Similarly, from $\triangle CAB$,

$$\frac{CG}{AG} = \frac{CF}{BF}$$

$$\text{i.e., } \frac{AG}{GC} = \frac{BF}{FC} \quad (2)$$

Therefore, from (1) and (2),

$$\frac{AE}{ED} = \frac{BF}{FC}$$

Example 3 : In Fig. 6.16, $\frac{PS}{SQ} = \frac{PT}{TR}$ and $\angle PST = \angle PRQ$. Prove that PQR is an isosceles triangle.

Solution : It is given that $\frac{PS}{SQ} = \frac{PT}{TR}$.

So, $ST \parallel QR$ (Theorem 6.2)

Therefore, $\angle PST = \angle PRQ$ (Corresponding angles) (1)

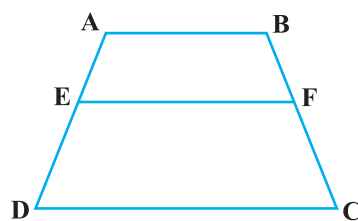


Fig. 6.14

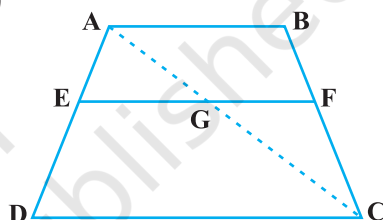


Fig. 6.15

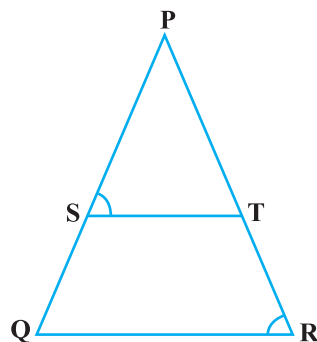


Fig. 6.16

Now, $PQ \parallel EF$ and $\triangle ABC \cong \triangle DPQ$ (How ?)

So, $\angle A = \angle D$, $\angle B = \angle P$ and $\angle C = \angle Q$

Therefore, $\triangle ABC \sim \triangle DEF$ (Why?)

We now take some examples to illustrate the use of these criteria.

Example 4 : In Fig. 6.29, if $PQ \parallel RS$, prove that $\triangle POQ \sim \triangle SOR$.

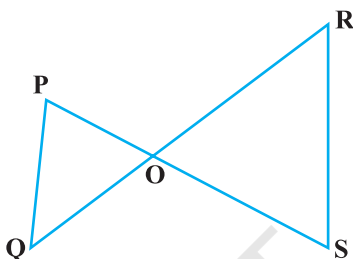


Fig. 6.29

Solution :

$PQ \parallel RS$ (Given)

So, $\angle P = \angle S$ (Alternate angles)

and $\angle Q = \angle R$

Also, $\angle POQ = \angle SOR$ (Vertically opposite angles)

Therefore, $\triangle POQ \sim \triangle SOR$ (AAA similarity criterion)

Example 5 : Observe Fig. 6.30 and then find $\angle P$.

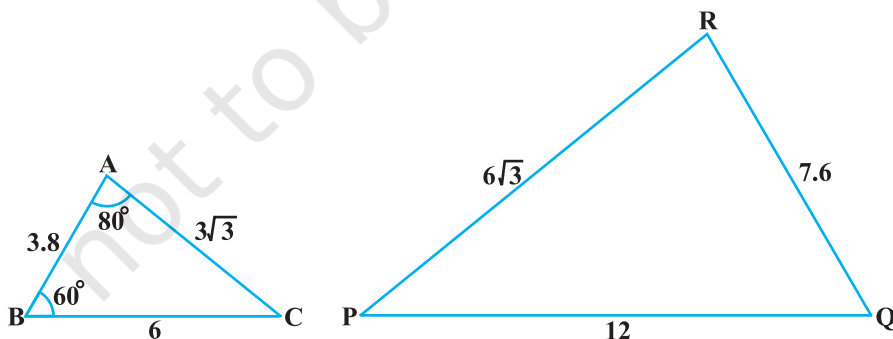


Fig. 6.30

Solution : In $\triangle ABC$ and $\triangle PQR$,

$$\frac{AB}{RQ} = \frac{3.8}{7.6} = \frac{1}{2}, \frac{BC}{QP} = \frac{6}{12} = \frac{1}{2} \text{ and } \frac{CA}{PR} = \frac{3\sqrt{3}}{6\sqrt{3}} = \frac{1}{2}$$

That is, $\frac{AB}{RQ} = \frac{BC}{QP} = \frac{CA}{PR}$

So, $\triangle ABC \sim \triangle RQP$ (SSS similarity)

Therefore, $\angle C = \angle P$ (Corresponding angles of similar triangles)

But $\angle C = 180^\circ - \angle A - \angle B$ (Angle sum property)

$$= 180^\circ - 80^\circ - 60^\circ = 40^\circ$$

So, $\angle P = 40^\circ$

Example 6 : In Fig. 6.31,

$$OA \cdot OB = OC \cdot OD.$$

Show that $\angle A = \angle C$ and $\angle B = \angle D$.

Solution : $OA \cdot OB = OC \cdot OD$ (Given)

So, $\frac{OA}{OC} = \frac{OD}{OB}$ (1)

Also, we have $\angle AOD = \angle COB$ (Vertically opposite angles) (2)

Therefore, from (1) and (2), $\triangle AOD \sim \triangle COB$ (SAS similarity criterion)

So, $\angle A = \angle C$ and $\angle D = \angle B$ (Corresponding angles of similar triangles)

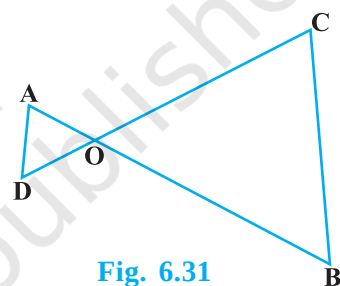


Fig. 6.31

Example 7 : A girl of height 90 cm is walking away from the base of a lamp-post at a speed of 1.2 m/s. If the lamp is 3.6 m above the ground, find the length of her shadow after 4 seconds.

Solution : Let AB denote the lamp-post and CD the girl after walking for 4 seconds away from the lamp-post (see Fig. 6.32).

From the figure, you can see that DE is the shadow of the girl. Let DE be x metres.

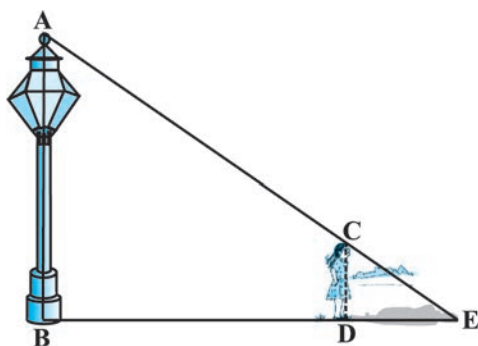


Fig. 6.32

Now, $BD = 1.2 \text{ m} \times 4 = 4.8 \text{ m}$.

Note that in $\triangle ABE$ and $\triangle CDE$,

$\angle B = \angle D$ (Each is of 90° because lamp-post as well as the girl are standing vertical to the ground)

and

$\angle E = \angle E$

(Same angle)

So,

$\triangle ABE \sim \triangle CDE$

(AA similarity criterion)

Therefore,

$$\frac{BE}{DE} = \frac{AB}{CD}$$

i.e.,

$$\frac{4.8 + x}{x} = \frac{3.6}{0.9}$$

$$(90 \text{ cm} = \frac{90}{100} \text{ m} = 0.9 \text{ m})$$

i.e.,

$$4.8 + x = 4x$$

i.e.,

$$3x = 4.8$$

i.e.,

$$x = 1.6$$

So, the shadow of the girl after walking for 4 seconds is 1.6 m long.

Example 8 : In Fig. 6.33, CM and RN are respectively the medians of $\triangle ABC$ and $\triangle PQR$. If $\triangle ABC \sim \triangle PQR$, prove that :

(i) $\triangle AMC \sim \triangle PNR$

(ii) $\frac{CM}{RN} = \frac{AB}{PQ}$

(iii) $\triangle CMB \sim \triangle RNQ$

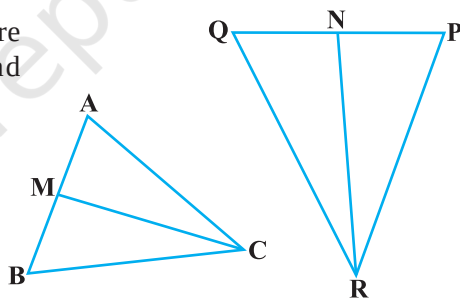


Fig. 6.33

(Given)

Solution : (i) $\triangle ABC \sim \triangle PQR$

So,

$$\frac{AB}{PQ} = \frac{BC}{QR} = \frac{CA}{RP} \quad (1)$$

and

$$\angle A = \angle P, \angle B = \angle Q \text{ and } \angle C = \angle R \quad (2)$$

But

$$AB = 2 \text{ AM and } PQ = 2 \text{ PN}$$

(As CM and RN are medians)

So, from (1),

$$\frac{2 \text{ AM}}{2 \text{ PN}} = \frac{CA}{RP}$$

8. If in two triangles, corresponding sides are in the same ratio, then their corresponding angles are equal and hence the triangles are similar (SSS similarity criterion).
9. If one angle of a triangle is equal to one angle of another triangle and the sides including these angles are in the same ratio (proportional), then the triangles are similar (SAS similarity criterion).

A NOTE TO THE READER

If in two right triangles, hypotenuse and one side of one triangle are proportional to the hypotenuse and one side of the other triangle, then the two triangles are similar. This may be referred to as the RHS Similarity Criterion.

If you use this criterion in Example 2, Chapter 8, the proof will become simpler.